the confusions that could be caused by someone with no experience would be to measure widths, say, by the angle subtended at the eye, and measure depth in a different way, like the strain on the muscles needed to focus them, so that the depths would be measured in feet and the widths in meters. Then one would get an enormously complicated mess of equations in making transformations such as (5.2), and would not be able to see the clarity and simplicity of the thing for a very simple technical reason, that the same thing is being measured in two different units. Now in Eqs. (5.1) and (5.3) nature is telling us that time and space are equivalent; time becomes space; *they should be measured in the same units*. What distance is a "second"? It is easy to figure out from (5.3) what it is. It is $3 \times 10^8$ meters, *the distance that light would go in one second*. In other words, if we were to measure all distances and times in the same units, seconds, then our unit of distance would be $3 \times 10^8$ meters, and the equations would be simpler. Or another way that we could make the units equal is to measure time in meters. What is a meter of time? A meter of time is the time it takes for light to go one meter, and is therefore $\frac{1}{3} \times 10^{-8}$ sec, or 3.3 billionths of a second! We would like, in other words, to put all our equations in a system of units in which $c = 1$. If time and space are measured in the same units, as suggested, then the equations are obviously much simplified. They are

$$\begin{aligned} x' &= \frac{x - ut}{\sqrt{1 - u^2}}, \\ y' &= y, \\ z' &= z, \\ t' &= \frac{t - ux}{\sqrt{1 - u^2}}, \end{aligned} \tag{5.4}$$

$$t'^2 - x'^2 - y'^2 - z'^2 = t^2 - x^2 - y^2 - z^2. \tag{5.5}$$

If we are ever unsure or "frightened" that after we have this system with $c = 1$ we shall never be able to get our equations right again, the answer is quite the opposite. It is much easier to remember